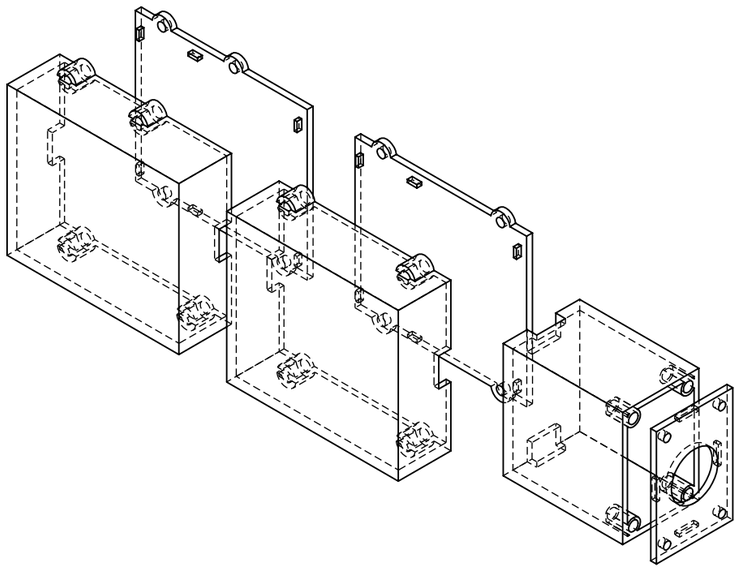
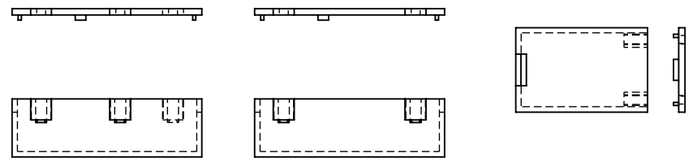
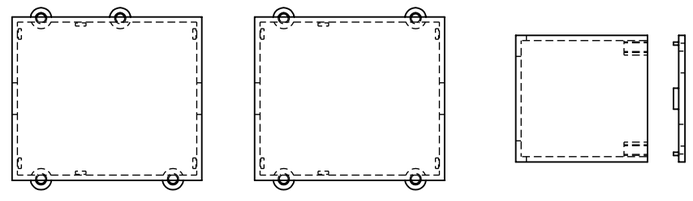
V-PIPE ROBOT - 3D PRINTABLE ENCLOSURE SPLIT DRAWING

Six printed parts | A3 exploded arrangement | Units: mm | Scale: as shown







E-01 POWER TOP COVER 72x62x22 t=2.0 QTY 1

E-02 POWER BASE 72x62 t=2.4 QTY 1

E-03 CONTROL TOP COVER 72x62x22 t=2.0 QTY 1

E-04 CONTROL BASE 72x62 t=2.4 QTY 1

E-05 CAMERA BODY 50x48x32 t=2.0 QTY 1

E-06 CAMERA BEZEL 48x32x2.4 / D19.5 QTY 1

FASTENERS: 8x M3x10 LOW BUTTON + inserts; 4x M2.5x8 + inserts

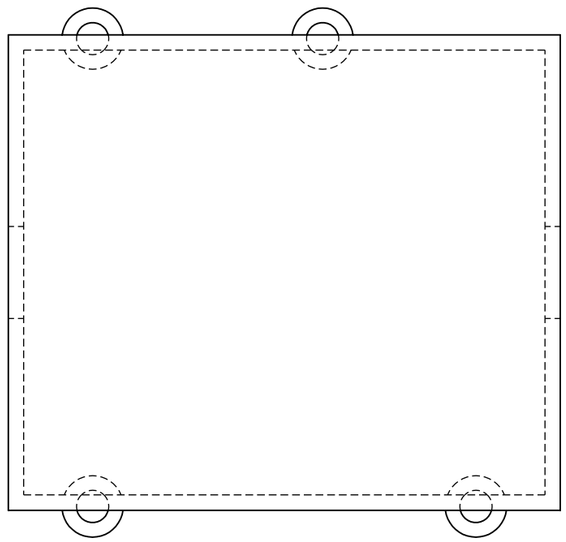
PRINT: PETG / FDM, 0.20 layer, 5 walls, 30-40% infill; covers roof-down

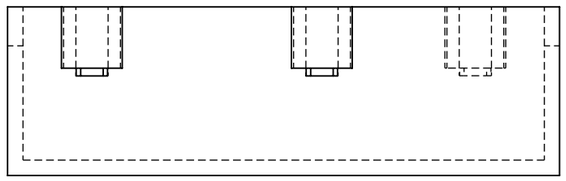
LOCATING TABS: bays 0.30 clearance; camera 0.50. See E-01 to E-06 sheets.

PROTOTYPE - VERIFY INSERT PILOTS ON A COUPON. DETAILS ON SHEETS 2-7.

E-01 | Power bay top cover, 2.0 mm wall

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1





Body: 72 x 62 x 22; cavity 68 x 58 x 20; wall/roof 2.0.

Overall width with fastening bosses/ears: 69.0.

4 x D4.2 through pilots in D8 x 8 bosses; M3 heat-set inserts.

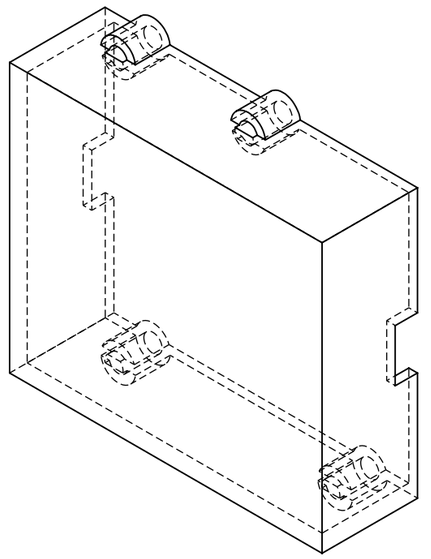
Hole coordinates relative to body centre (X,Y):

(-25,-30.5), (25,-30.5), (-25,30.5), (5,30.5).

2 x cable slots at end-wall rim: 12 wide x 5.1 clear height.

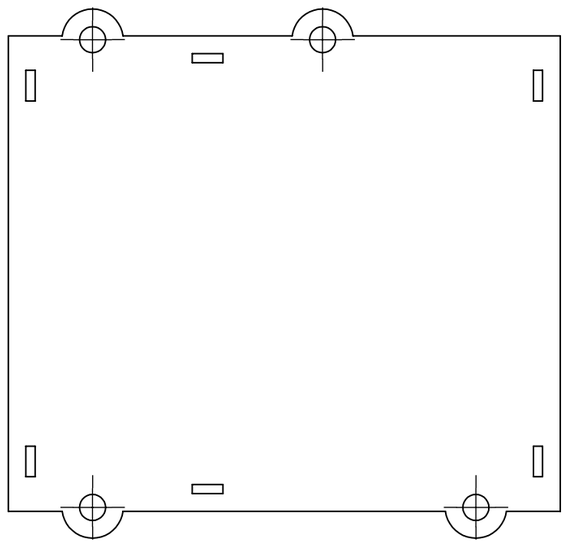
4 x M3x10 LOW BUTTON heads: D<=5.7, H<=1.7; M3 inserts L5-6.

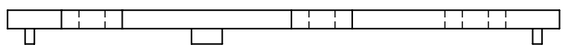
Print roof on bed, opening upward; support boss starts locally.



E-02 | Power bay base plate, 2.4 mm

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1





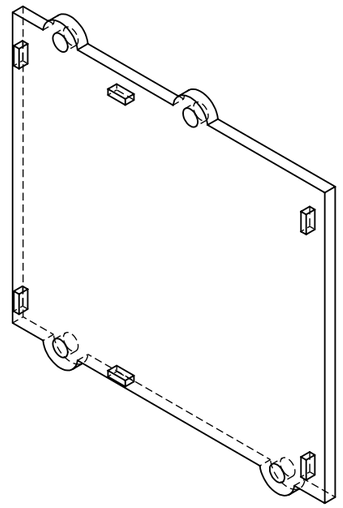


Plate: 72 x 62 x 2.4; overall height with tabs 4.4.

Overall width with fastening bosses/ears: 69.0.

4 x D3.4 clearance holes. Do not tap the printed plate.

Hole coordinates relative to body centre (X,Y):

(-25,-30.5), (25,-30.5), (-25,30.5), (5,30.5).

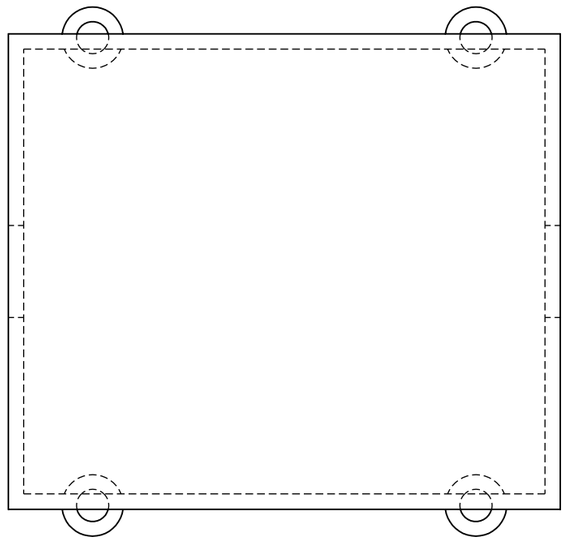
6 locating tabs: 2.0 engagement; 0.30 clearance to cavity.

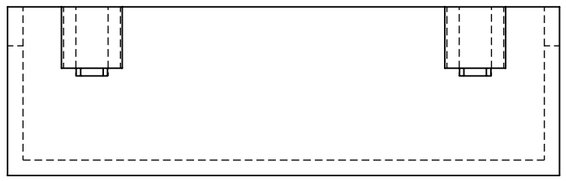
4 x M3x10 LOW BUTTON heads: D<=5.7, H<=1.7; M3 inserts L5-6.

Print flat underside on bed; locating tabs upward.

E-03 | Control bay top cover, 2.0 mm wall

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1





Body: 72 x 62 x 22; cavity 68 x 58 x 20; wall/roof 2.0.

Overall width with fastening bosses/ears: 69.0.

4 x D4.2 through pilots in D8 x 8 bosses; M3 heat-set inserts.

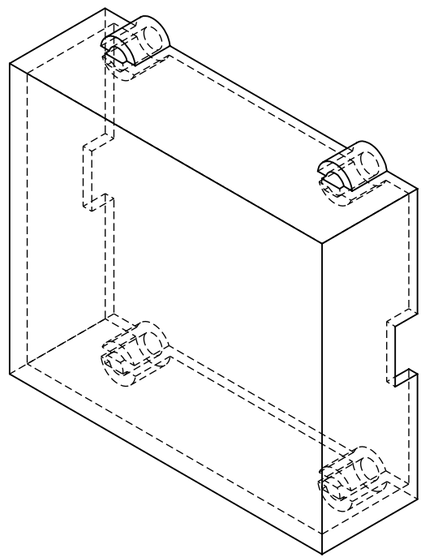
Hole coordinates relative to body centre (X,Y):

(-25,-30.5), (25,-30.5), (-25,30.5), (25,30.5).

2 x cable slots at end-wall rim: 12 wide x 5.1 clear height.

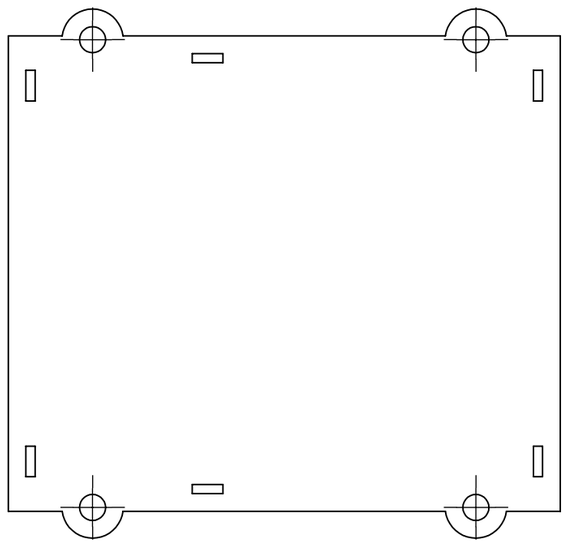
4 x M3x10 LOW BUTTON heads: D<=5.7, H<=1.7; M3 inserts L5-6.

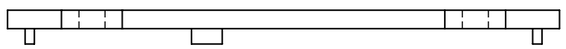
Print roof on bed, opening upward; support boss starts locally.



E-04 | Control bay base plate, 2.4 mm

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1





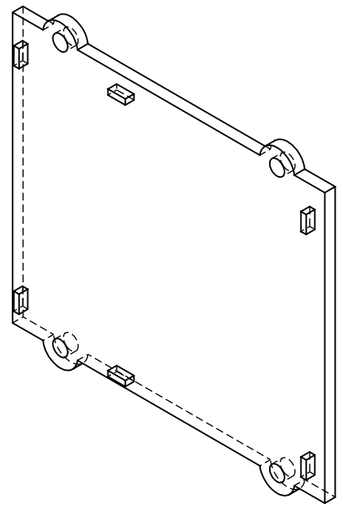


Plate: 72 x 62 x 2.4; overall height with tabs 4.4.

Overall width with fastening bosses/ears: 69.0.

4 x D3.4 clearance holes. Do not tap the printed plate.

Hole coordinates relative to body centre (X,Y):

(-25,-30.5), (25,-30.5), (-25,30.5), (25,30.5).

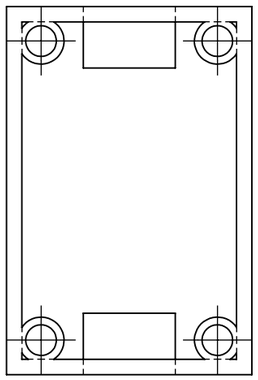
6 locating tabs: 2.0 engagement; 0.30 clearance to cavity.

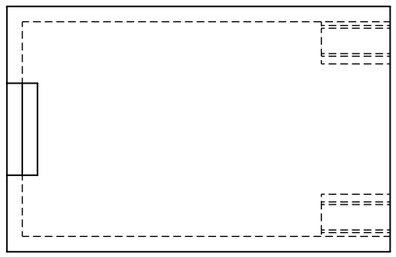
4 x M3x10 LOW BUTTON heads: D<=5.7, H<=1.7; M3 inserts L5-6.

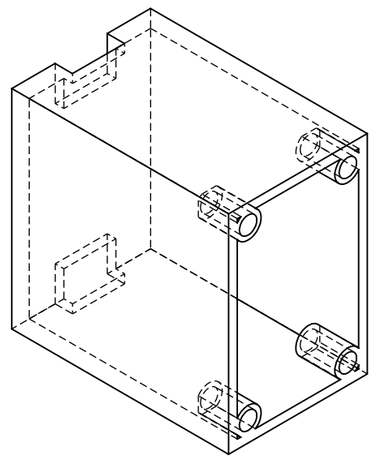
Print flat underside on bed; locating tabs upward.

E-05 | Camera housing body, rear-wall print orientation

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1







Body: 50 x 48 x 32; wall 2.0; front open.

4 holes along X; (Y,Z) relative to optical axis:

(-19.5,-11.5), (-19.5,11.5), (19.5,-11.5), (19.5,11.5).

4 x D4.0 pilots in D6 x 9 bosses; M2.5 heat-set inserts.

2 rear-edge cable slots: 9 wide x 12 high; add split grommets.

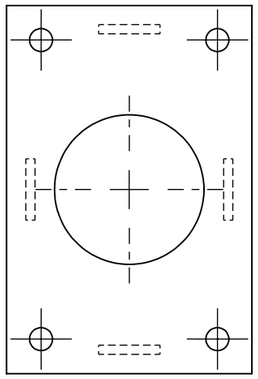
Print rear wall down; support the two cable-entry bridges.

Camera/lens measured mounting details remain to be fitted.

Pilot sizes must match the purchased insert; print a test first.

E-06 | Camera front bezel, D19.5 aperture

PETG / FDM | QTY 1 | mm | Orthographic scale 1.25:1 | ISO 1:1





Front plate: 48 x 32 x 2.4; D19.5 lens aperture.

4 holes along X; (Y,Z) relative to optical axis:

(-19.5,-11.5), (-19.5,11.5), (19.5,-11.5), (19.5,11.5).

4 x D3.0 screw clearance; tabs engage 2.0 with 0.50 clearance.

Locate by four tabs, then close with 4 x M2.5x8 screws.

Print front face down; tabs grow upward.

Camera/lens measured mounting details remain to be fitted.

Pilot sizes must match the purchased insert; print a test first.

